# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-GEODESY-EN · v2.2 · 2026-11-02 |
| Package | star-geodesy |
| Version | 0.1.1 |
| Capability | star.geodesy.ecef |
| Maturity | Beta (VERIFIED) |
| Python | ⩾3.9 |
| Source commit | dc09f61 |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_geodesy |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_geodesy-0.1.1-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_geodesy-0.1.1-py3-none-any.whl | 5d8469c8e943604b4560a2dbcc2cd03085cb83c77268658de33a1f6df31903aa |
| star_geodesy.py | b64fb8d6cbe9e2df7fec7f89298e3ffe4534e3ae5960dc4d050a1584739261b4 |
| star_geodesy-0.1.1-source.zip | ae31c3d501a9c67d3430069c1acfe46c7fdf58a0d769898d41b97a0df65718ac |

# 5. What it does, and its limits

WGS-84 geodetic coordinates to and from ECEF. Standard library only.

Other datums, geoid heights, geodesic distances (see XC-003), any flight use.

- `geodetic_to_ecef(lat_deg, lon_deg, h_m)` → `(x, y, z)` in metres (WGS-84, NIMA TR8350.2 constants).
- `ecef_to_geodetic(x, y, z)` → `(lat_deg, lon_deg, h_m)`, iterative to 1e-14 rad at ANY altitude; RuntimeError if it does not converge; the forward transform of the result closes to < 1 micrometre.
- Invalid input raises `ValueError` (|lat| > 90, Earth's centre).

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

# 7. Citation and bill of materials

```
@software{cavazzini_star_geodesy_0_1_1,
  author  = {Cavazzini, Andrea},
  title   = {{star-geodesy}: WGS-84 geodetic to/from ECEF with exact inversion at any
altitude},
```

```
  version = {0.1.1},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_geodesy/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts